

NCSU's Virtual Computing Lab: A Cloud Computing Solution



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North Carolina State University's Virtual Computing Lab cost-effectively and efficiently delivers diverse computing resources over the Internet to more than 30,000 students and faculty.

University IT organizations face many challenging questions: Can we meet the costs and diverse requirements of the student computer lab? Do we need on-demand desktops? Are support costs escalating due to conflicts among the dozens of applications installed on the lab computers? Do off-campus students have sufficient access to computing resources? Can we afford all the software students require? Are there compatibility issues with student-owned computers? When should we install a newly released version of an important application demanded by faculty? Do we need reconfigurable high-performance computing (HPC) resources? Are computing lab resources idle?

At North Carolina State University, we struggled with these and many other related issues. Getting off to a good start looked difficult, and doing it with minimal resources didn't make the situation any easier.

VIRTUAL COMPUTING LAB

In 2003, NCSU faculty and staff considered the concept of delivering computing lab applications to remote users but couldn't identify the right set of tools to implement it. We installed and evaluated several commercial packages, but none of them combined the required performance with the necessary affordability.

Coincidentally, at the same time NCSU began a centralized HPC service (<http://hpc.ncsu.edu>). A primary HPC resource was a Linux cluster using IBM BladeCenter hardware managed with IBM's freely available Extreme Cluster Administration Toolkit (xCAT). The experience with this platform made it clear that it could also be used to deliver arbitrary images loaded on a blade server on demand.

This realization led to the birth of NCSU's Virtual Computing Laboratory (<http://vcl.ncsu.edu>). A central VCL requirement was affordability. Our goal was to use standard industrial components, along with a combina-

tion of open source and proprietary software (including VCL code developed in Perl and PHP at NCSU), to flexibly and economically meet the lab's needs. Economy of scale and operation, simplicity, ease of access and usability, ease of maintenance, and automation were the keys to providing the needed computing services efficiently and effectively. An important side benefit was substantial conservation of electrical energy.

VCL OPERATION

NCSU's central HPC group partnered with the university's College of Engineering IT group to develop a Web interface that students could use to request access to specific *images*. These are software stacks that range from hypervisor, to operating system, to application, to service access methods. We used xCAT to load the requested "bare-metal" image onto a blade server. Neither group had specific funding for the project, just a problem that needed to

be solved: application delivery to on- and off-campus students, and onto student-owned computers.

The initial design had to be simple. We used a MySQL database to track each server's state, maintain information about each image, and implement a privilege tree. A Web interface, driven by PHP, authenticated students, displayed a list of applications they were authorized to use, and let them reserve use of an application either immediately or sometime in the future for a specified length of time. The range of future times and length of reservation were customizable and could differ based on the user.

Today, when an authenticated user makes a reservation request on the Web interface, the Web portal schedules the request.

While originally VCL was bare-metal-oriented, today it loads either a VMware-based image or a bare-metal image. The VCL system processes the request. If it doesn't find an available real or virtual server with the desired image already loaded, it selects any available server meeting the specifications required for that image, and xCAT, or the appropriate VM loader, dynamically loads the desired image. If all servers are busy, the Web interface presents the student with a grid showing future times when they will be available. Each image is equipped with clients that can access the NCSU enterprise storage data network, any network-accessible storage, and private storage on the end user's access device.

We incorporated security into many aspects of the VCL design from the beginning (<http://vcl.ncsu.edu/papers-publications>). One important feature is that the VCL system re-images the blade servers for each user session, thereby providing users with administrator/root privileges, if desired. Users can easily modify images, but that doesn't affect the

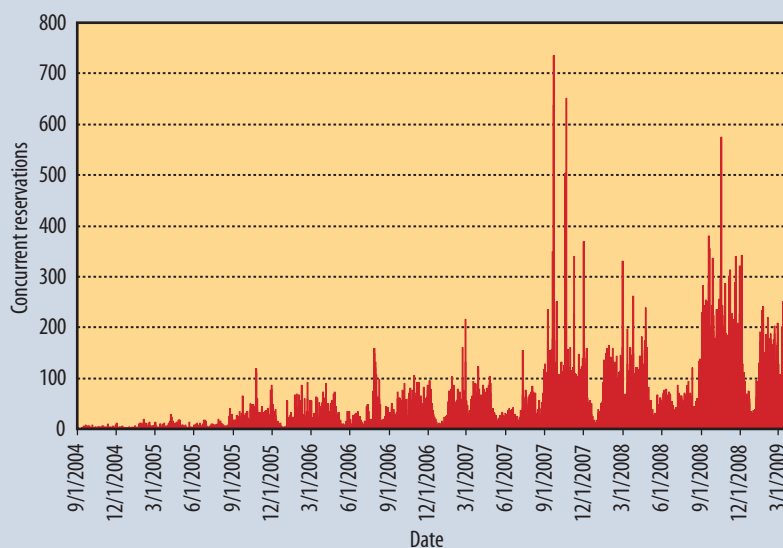


Figure 1. Virtual Computing Lab reservations from September 2004, when the VCL project began, through February 2009. While there were barely 700 reservations during the pilot semester, there are now more than 80,000 per semester.

image library and therefore subsequent users.

VCL GROWTH

We piloted the initial VCL system with a few engineering course sections during the fall 2004 semester. The system delivered Windows applications to users via the Remote Desktop Protocol (RDP) and Linux applications to users via Secure Shell (SSH) and the X Window System Protocol. Other access methods were also used, including Web services.

Most NCSU engineering students have laptop computers. With RDP and Secure Shell/X Window, VCL can deliver both Windows and Linux applications to Windows PC, Mac, and Linux laptops. A broadband Internet connection is a practical requirement.

Feedback from users—including many outside the country—was positive, and at the end of the pilot semester a software vendor provided the impetus for a second college to become involved with VCL. The vendor altered the license terms from allowing distribution of its application to students to requiring that it be run only on university-owned computers. This was a relatively heavily

used application, and students were accustomed to using it on their own computers—an inexpensive student version of the software had limitations that often rendered it unsuitable.

Requiring that students go to a campus computer lab to use the application would have provoked an uproar. Fortunately, providing access to the application via VCL satisfied the license terms while allowing students to continue to access the application from their own computers.

Since then, the number of users has increased each semester. We now serve more than 30,000 students and faculty from a dozen or so campuses. There are also numerous pilots in several states, as well as abroad. We have recorded more than 400,000 individual sessions since the project began. While there were barely 700 reservations during the pilot semester, there are now more than 80,000 per semester.

VCL use continues to increase, as Figure 1 shows, confirming that investment in its resources was worthwhile. With the VCL/HPC cloud now serving more than 7 million HPC CPU hours per year, our approach supports large-scale cloud computing at cents to tens of cents per CPU

hour depending on the type of services provided.

SYSTEM UPGRADES

A few years ago, a computer science researcher wanted to modify the kernel on the HPC Linux cluster to try to improve performance. The central HPC Linux cluster is a production resource used by researchers from across the university. Running production work on an experimentally tweaked kernel isn't the level of service central IT wants to deliver, but satisfying CS researchers' needs is also important.

applications, we extended VCL to use virtual (hypervised) servers in addition to physical servers. The system creates virtual servers by loading a host virtual server image onto a blade server. VCL manages guest virtual machines by issuing commands to the hypervisor—which is primarily VMware at this time.

ECONOMIC BENEFITS

While we have had to expand hardware to accommodate the increased use of VCL, it hasn't been necessary to increase staff to support the system. In NCSU's experience, a small staff

control software adjusts virtual local area network (VLAN) settings on the chassis Ethernet switches to connect the servers to private HPC networks instead of the networks used for individual application delivery to users connecting through the Internet.

The master HPC login node image includes components of the HPC scheduler—currently Platform Computing's Load Sharing Facility (LSF). Each HPC client image has access to a large amount of scratch storage (in terabytes), as well as end-user home directories and HPC backup storage. When an HPC image is loaded, the master scheduler recognizes it and begins to assign it work. To transition out of the Linux HPC cluster mode, our staff closes blades to new HPC jobs and reassigns them to be loaded with individual VCL images once any running jobs complete. In this way, the blade servers are in use many more hours per year than are traditional student lab computers, and the added availability allows much more HPC research computing than would otherwise be possible.

This amortization of VCL cost over many additional uses and users substantially decreases operation costs. The number of uses of a particular server can be increased in two ways relevant to VCL: time and virtualization. First, we keep the server busy as many hours per year as possible. Different applications and users will have different usage patterns and length of use. Having a mix of usage types and patterns reduces time gaps to be filled and keeps servers busy. Second, we use one physical server as multiple virtual servers to serve multiple uses concurrently. Appropriately assigning applications to servers—physical or virtual—is important for both quality of service and cost-efficiency.

Today the image creator chooses between virtual and dedicated server use in the image metadata: how much memory, how much processing power, and so on. Ideally these would be dynamic decisions based not only

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VCL allowed researchers to produce an image of the custom kernel, and our developers extended the VCL system so that it could load an image onto multiple servers with a single reservation. This ability has proven popular not only for researchers wanting a cluster on demand, or "subcloud," but also for users seeking complex environments—for example, a database server, application server, and Web server running together.

Another popular extension to the VCL system has been the addition of block reservations designed to support use by classes. A course instructor makes a reservation—say, for 30 seats 10:00 to 10:50 a.m. every Monday and Friday for the semester—for a particular application. Students visit the VCL portal and have access to the already loaded instances of the application image based on their inclusion in the class roll.

As the range of applications delivered by VCL expanded, we found that many didn't need the full resources of a physical server. With the increasing availability and use of multicore processors, more applications are falling into this category. To achieve efficient hardware utilization with these

can provide a diverse set of applications and services using VCL—many of which wouldn't have been feasible without the system.

While VCL currently supports more than 600 production images, it centrally maintains only 10 to 20. These are used as baseline templates that can then be expanded into end-user production images, thereby distributing the cost of image development and maintenance.

The implication of this experience is that scale and utilization level are important for providing cost-efficient VCL/HPC services. At small scale, staff cost dominates—even with a small staff. As scale increases, staff cost is relatively constant and is amortized over more uses, and cost primarily depends on the hardware resource cost and utilization level. This points to maximizing use of the hardware infrastructure.

When students leave for the summer or breaks and individual use diminishes, we use the VCL hardware resources to expand HPC services. This transition is accomplished without any physical changes. Under VCL control, xCAT loads an HPC compute image onto idle nodes, and the VCL

on the image, but also on the particular user requesting the image and his or her usage profile—for example, power user versus novice user.

NCSU realized that there are numerous desirable extensions of VCL capability, but that they greatly exceed the capacity of its available development resources. Consequently, the university has donated the VCL code to the Apache Software Foundation, where VCL has been accepted as an incubator project (<http://cwiki.apache.org/VCL>). Through Apache, a community of developers, including some from NCSU, can continue to extend and enhance VCL while keeping the code freely available.

Seldom considered in the overall cost calculations are the initial cost and lifetime of computers used directly. Often these are overspecified or replaced while still in good working order to be able to run occasional high-end applications. With the VCL, the user's computer—running RDP, X Window, or Web service clients—

serves only as an access device, with minimal requirements. At the same time, VCL's virtual aspect provides anytime, anywhere delivery that now enables the delivery of services such as compute-intensive courses via distance education, which opens course work to many.

The delivery of many diverse computing services over the Internet, with flexible provisioning, has led to much greater efficiency, substantial cost savings, and many ways to enable and empower end users. NCSU's own experience with cloud computing, through its Virtual Computing Lab, indicates that this approach would be beneficial to a much wider audience. **E**

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